MEM1: Learning to Synergize Memory and Reasoning for Efficient Long-Horizon Agents

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TL;DR

We introduce MEM1, an end-to-end reinforcement learning framework that enables agents to operate with constant memory across long-horizon tasks. At each turn, MEM1 updates a compact shared internal state that jointly supports memory consolidation and reasoning. This state integrates prior memory with new observations from the environment while strategically discarding irrelevant or redundant information.

Problem Description

web search household robot coding agent

Given a task, an agent iteratively uses external tools and gathers information from the environment until the task is completed.

Motivating Question

How can we enable agents to perform complex tasks that involve lots of interactions with the environment?

- Challenge 1: Growing inference cost and memory usage O(N2) for compute and O(N) for memory
- Challenge 2: Overloaded context leads to degraded performance

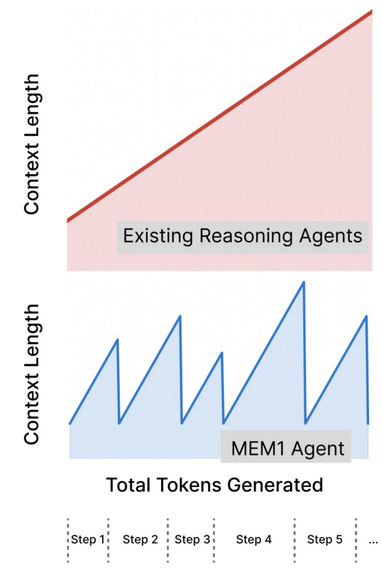
The Gap in Literature

- Existing works have focused on using external memory, but can context management be thought as a reasoning process?
- If so, can the agent be trained to do well in context management?

MEM1

Key Observation

During reasoning, it is common and helpful for the model to summarize key information from previously encountered content.



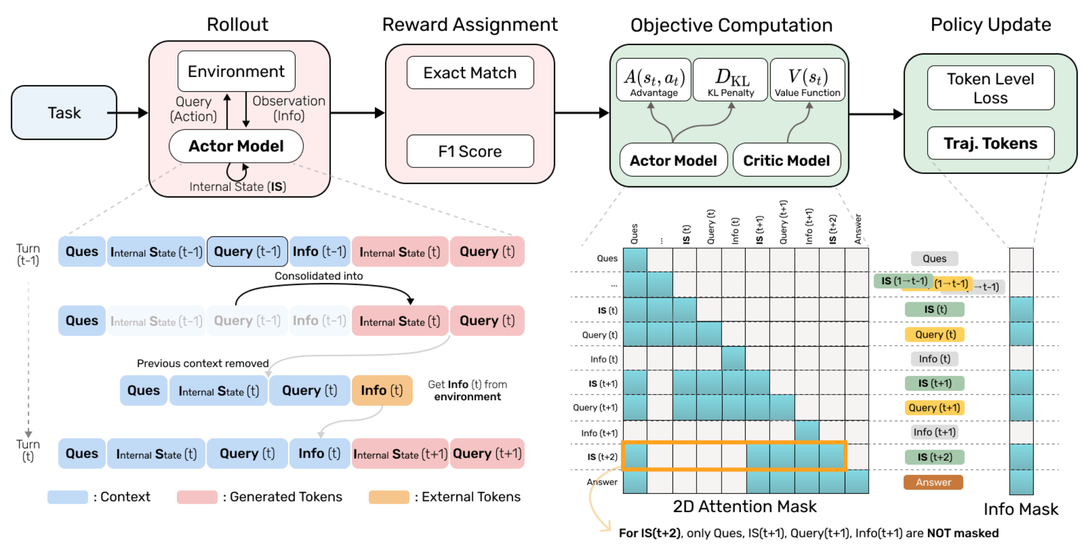
Proposal

Introduce an internal state that integrates reasoning and memory. At each step, the model updates this internal state and discards all previous information.

Advantages

1. Enable near-constant memory
2. Support end-to-end optimization

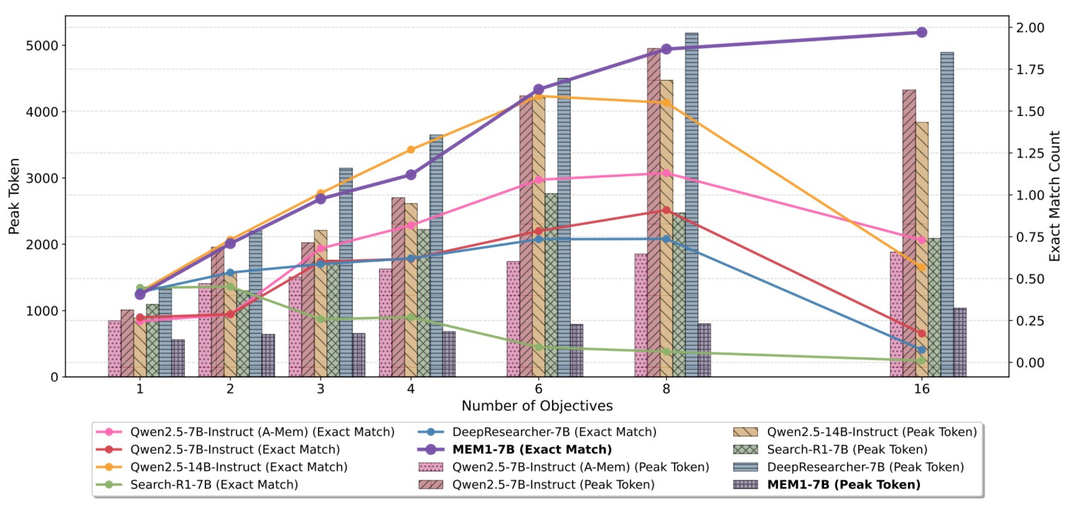
Practical Implementation



1. During rollout, context is modified at each turn
2. At the start of each turn, the agent first consolidates context into its reasoning (i.e., internal state)
3. At the end of the turn, all information before the consolidation is truncated
4. All contexts are trained based on the reward of the trajectory.

Experiments

Perf. as no. objectives increases



MEM1 retains decent performance even with a large number of objectives to deal with.

Table 1: Comparison of models on multi-objective multi-hop QA tasks. Arrows indicate the desired directions. Numbers in red indicate collapsed model behavior (extremely low performance). (truncate) means using MEM1's prompt and rollout pipeline. (A-MEM) means using MEM1's prompt and rollout pipeline with A-Mem's external memory module (Xu et al., 2025). MEM1-QA means MEM1 trained on 2-objective QA task.

Model	2-Objective				8-Objective				16-Objective			
	EM ↑	F1 ↑	Peak ($\times 10^2$) ↓	Time (s) ↓	EM ↑	F1 ↑	Peak ($\times 10^2$) ↓	Time (s) ↓	EM ↑	F1 ↑	Peak ($\times 10^2$) ↓	Time (s) ↓
Qwen2.5-14B-Inst	**0.732**	**0.902**	15.6±0.19	5.49 ± 0.16	1.55	1.87	44.7 ± 0.37	16.2 ± 0.27	0.567	0.703	38.4±0.71	29.7±0.75
Qwen2.5-7B-Inst	0.268	0.366	19.6±0.33	4.60±0.08	0.87	1.10	49.5±0.40	13.9±0.18	0.165	0.213	43.3±0.62	15.5±0.23
Qwen2.5-7B-Inst (A-MEM)	0.286	0.371	14.1±0.10	24.6±0.51	1.13	1.43	18.6±0.10	53.7±1.26	0.730	0.961	18.8±0.14	91.2±2.44
Qwen2.5-7B-Inst (truncate)	0.262	0.336	8.28±0.06	5.89±0.16	0.97	1.23	11.8±0.10	11.9±0.20	0.396	0.497	13.3±0.16	22.1±0.60
Search-R1 (original)	0.452	0.531	13.0±0.08	4.09 ± 0.23	0.064	0.08	24.7 ± 0.19	**4.25±0.16**	0.009	0.011	20.9±0.03	**4.75±0.18**
Search-R1 (trained on 2-obj task)	0.544	0.646	13.68±0.23	10.60 ± 0.32	0.471	0.575	22.48±0.47	19.60±0.50	0.520	0.647	24.8±0.57	23.35±0.77
Search-R1 (trained on 2-obj task + truncate)	0.446	0.546	**6.12±0.04**	17.5 ± 0.22	0.162	0.204	**4.95±0.02**	21.5±0.18	0.091	0.107	**5.28±0.03**	24.6±0.22
DeepResearcher	0.536	0.650	22.0±0.43	**4.01±0.07**	0.73	0.90	51.8±0.35	11.3±0.14	0.071	0.106	48.9±0.66	15.8±0.19
MEM1-QA	0.709	0.838	**6.40±0.02**	6.49 ± 0.07	**1.87**	**2.31**	8.01±0.06	**8.68±0.12**	**1.97**	**2.39**	**10.4±0.09**	**8.70±0.12**

Table 2: The experimental results for WebShop. For a fair comparison, we do not report GPT's inference time. For Agent-R, scores are taken from the original paper, as the model is closed source. MEM1-WebShop means MEM1 trained on WebShop environment.

Model	Avg Final Reward ↑	Peak Token ($\times 10^3$) ↓	Dependency ($\times 10^6$) ↓	Inference Time Per Traj (s) ↓
GPT-4o	25.48	5.30 ± 1.23	3.99 ± 1.16	*N/A*
GPT-4o (truncate)	13.82	0.99 ± 0.99	0.81 ± 0.23	*N/A*
GPT-4o (A-MEM)	24.50	1.84 ± 0.06	0.31 ± 0.11	*N/A*
Qwen2.5-7B-Instruct	18.42	5.64 ± 1.34	3.38 ± 0.89	12.31 ± 1.82
Qwen2.5-14B-Instruct	12.34	5.44 ± 0.92	3.30 ± 0.61	18.17 ± 2.32
Agent-FLAN-7B	40.35	3.37 ± 1.12	2.18 ± 1.62	9.95 ± 6.19
Agent-R-8B	63.91	*N/A*	*N/A*	*N/A*
AgentLM-7B	63.60	2.24 ± 0.40	0.28 ± 0.07	3.91 ± 1.07
AgentLM-13B	70.80	2.36 ± 0.46	0.30 ± 0.08	5.23 ± 1.59
MEM1-WebShop	**70.87**	**0.81 ± 0.10**	**0.15 ± 0.16**	**2.61 ± 0.48**